INCH-POUND

MIL-L-00757B (AR)
16 May 1994
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MILITARY SPECIFICATION

LEAD STYPHNATE, NORMAL

This specification is approved for use by the US Army Armament, Munitions and Chemical Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-L-757. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of normal lead styphnate for use in primer compositions (Section 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA

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SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard for Testing Purposes

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129-1	-	Marking for Shipment and Storage Ammunition and Explosives
MIL-STD-650	-	Explosive: Sampling Inspection and Testing
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1218	-	ACS Chemicals
MIL-STD-2073-1	-	DOD Material Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS

49 CFR 100-180	-	Department of Transportation Rules and Regulations for the Transportation of Hazardous Materials
NBS Standard 136	-	National Bureau of Standard Potassium Dichromate

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DI PACK - 81059	-	Data Item Description
U.N. Document STSG/AC.10.11	-	United Nations Committee of Experts on the Transportation of Dangerous Goods
ICAO	-	International Civil Aviation Organization
IMDG Code	-	Intergovernmental Maritime Organization's International Maritime Dangerous Goods Code

(The Interstate Commerce Commission Regulation are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite, "49 CFR 100-180 (latest revision)".)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D2905	-	Statements on Number of Specimens Required to Determine the Average Quality of a Textile Material
ASTM E70	-	pH of Aqueous Solutions with Glass Electrode, Test for Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

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3.1.1 Appearance. Normal lead styphnate shall be free of visible impurities or foreign materials when tested as specified in 4.5.2.

3.1.2 Crystal form. Normal lead styphnate shall be crystalline. The longest dimension of any crystal shall not be greater than three times the next longest axis. Hexagonal plates shall be acceptable. The sample shall be tested as specified in 4.5.3.

3.1.3 Crystal dimensions. Unless otherwise specified in the contract or purchase order, the largest dimension of any normal lead styphnate crystal or aggregate of crystals shall not be greater than 0.5 mm when tested as specified in 4.5.4 (see 6.3).

3.2 Chemical and physical characteristics. The chemical and physical characteristics of normal lead styphnate shall conform to the requirements given in Table I when tested using the specified test method.

TABLE I. Physical and chemical characteristics.

Characteristics	Requirement	Test method
Granulation	100% through U.S. Std # 100 sieve	4.5.5
Apparent density, gm/cc	1.30 to 1.60	4.5.6
Acidity, pH	5.0 to 6.6	4.5.7
Material insoluble in ammonium acetate, max percent	0.3	4.5.8
Material soluble in ether, max percent	0.1	4.5.9
Lead, percent	43.2 to 44.3	4.5.10
Nitrogen, percent	8.87 to 9.07	4.5.11
Instantaneous flash point, °C (see 6.3)	310 to 330	4.5.12

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3.3 First article. When specified in the contract or purchase order document (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with the technical provisions herein (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspections requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The required inspections specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

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4.3 First article inspection.

4.3.1 Submission. When specified in the contract or purchase order, the contractor shall submit a first article sample consisting of one (1) pound of normal lead styphnate. The first article sample shall be obtained from the first production lot which has been produced by the contractor, using the same production processes, procedures and equipment as will be used in fulfilling the contract. All raw materials used in producing the normal lead styphnate shall be from the same sources as will be used in regular production.

4.3.2 Inspections to be performed. The first article sample may be subjected by the Government to any or all of the examinations or tests specified herein.

4.3.3 Rejection. If the sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

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TABLE II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Characteristics prior to packing		NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Appearance		3.1.1	4.5.2
	Crystal Form		3.1.3	4.5.3
	Crystal dimensions		3.2	4.5.4
	Granulation		3.2	4.5.5
	Apparent density		3.2	4.5.6
	Acidity		3.2	4.5.7
	Material insoluble in ammonium acetate		3.2	4.5.8
	Material soluble in ether		3.2	4.5.9
	Lead		3.2	4.5.10
	Nitrogen		3.2	4.5.11
	Instantaneous flash point		3.2	4.5.12

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. A lot shall consist of one or more batches not exceeding 500 pounds produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of normal lead styphnate that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The criteria and procedure for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable specification requirements utilizing as a minimum the conformance criteria specified herein.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.8). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Chemical and physical characteristics			NEXT HIGHER ASSEMBLY
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Appearance	4.4.3.2	3.1.1	4.5.2
102	Crystal form	4.4.3.2	3.1.2	4.5.3
103	Crystal dimensions	4.4.3.2	3.1.3	4.5.4
104	Granulation	4.4.3.3	3.2	4.5.5
105	Apparent density	4.4.3.2	3.2	4.5.6
106	Acidity, pH	4.4.3.2	3.2	4.5.7
107	Material insoluble in ammonium acetate	4.4.3.3	3.2	4.5.8
108	Material soluble in ether	4.4.3.3	3.2	4.5.9
109	Lead	4.4.3.3	3.2	4.5.10
110	Nitrogen	4.4.3.3	3.2	4.5.11
111	Instantaneous flash point	4.4.3.3	3.2	4.5.12
Minor	None defined			

NOTES:

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4.4.3 Testing. PRECAUTION: This specification covers sampling and testing of toxic and hazardous material. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing.

4.4.3.1 Sampling. Obtain two random 30 gm samples from each batch of normal lead styphnate being submitted for inclusion in a lot. The samples shall be obtained from the process vessel prior to packout using procedures described in ASTM-E300 for slurries.

4.4.3.2 Testing for batch acceptance. One sample obtained according to 4.4.3.1 from each batch shall be tested in accordance with applicable test methods for conformance with the requirements of appearance (see 3.1.1), crystal form (see 3.1.2), crystal dimensions (see 3.1.3), apparent density (see 3.2), and acidity (see 3.2). If any sample fails to meet any of these requirements, the batch represented by the sample shall be rejected. Rework of non-conforming batches is permitted.

4.4.3.3 Testing for lot acceptance. The second samples from the batches not rejected under the provisions of 4.4.3.2 shall be placed in a suitable container of sufficient volume to hold a composite sample from all batches in the lot. The composite sample shall be thoroughly mixed to assure homogeneity, and two sub-samples of approximately 30 gm each shall be obtained from this composite sample. Both sub-samples shall to tested in accordance with the applicable test methods of 4.5 (see 3.2, Table I) to determine conformance with the requirements of granulation, nitrogen content, instantaneous flash point, material insoluble in ammonium acetate, material soluble in ether and lead content. If either sub-sample fails to meet any of the test requirements, the lot represented by the composite sample shall be rejected.

4.4.4 Inspection equipment. For the performance of all test and examinations specified in 4.3 and 4.4, the equipment specified shall be employed. the contractor shall have the equipment available, utilize it in accordance with the test method, and is responsible to assure that it is properly calibrated. Government approval of all contractor designed inspection equipment is required prior to its use for acceptance testing (see 6.5).

4.5 Methods of inspection. All tests described in this section shall be performed using prescribed procedures for replicate determination given in standard analytical chemistry textbooks or ASTM D 2905. Unless otherwise specified herein, all

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chemicals shall be Reagent Grade or ACS Grade in accordance with MIL-STD-1218. If these grades are not available, all chemicals shall be the highest purity grade commercially available. See 6.6 for the use of equivalent test methods.

4.5.1 Preparation of dry sample. Transfer a portion of the wet sample containing about 10 gm of lead styphnate to a Buchner funnel containing a medium texture filter paper and apply suction until sample is almost dry. While working behind a safety shield, spread the sample over a watch glass. Dry the sample by placing the watch glass in a desiccator over drying agent for 24 hours or in an oven at 70° to 75°C until constant weight is obtained (this will take at least 2 hours). Use this dry sample for all determinations unless otherwise specified.

4.5.2 Appearance. Take a portion of the well stirred wet sample from 4.4.3.1 and spread over an area of approximately 4 sq. cm. on a glass microscope slide. Allow the sample to air dry. Examine the material microscopically for impurities under daylight illumination. Retain the slide for the determination of crystal form (see 4.5.3) and crystal dimensions (see 4.5.4).

4.5.3 Crystal form. Using the slide prepared in 4.5.2, observe the crystalline form (shape) under a microscope using a magnification of about 100X under reflected light, and optionally, take a photomicrograph at about 50X magnification.

4.5.4 Crystal dimensions. Using the slide prepared in 4.5.2, measure the longest dimension of the larger crystals observed when scanning the field with the aid of a mechanical stage. Use transmitted and reflected light and an ocular micrometer. Consider the longest dimension to be the distance between the most remote points of any one crystal.

4.5.5 Granulation. Transfer about 30 gm of the wet sample to a Buchner funnel and apply suction for 2-3 minutes. Weigh out a 10 gm portion of the damp sample and transfer it to a clean three inch US Standard No. 100 sieve which conforms to the requirements of RR-S-366. Carefully examine the sieve to be sure it is free of foreign material. Place the sieve in a porcelain evaporating dish which contains enough water to cover the wire screen of the sieve to a depth of about one half inch. Shake the sieve for a few minutes in such a manner that the lead styphnate is below the surface of the water and any material passing through the sieve is caught in the evaporating dish. When most of the lead styphnate has passed through the sieve, run a gentle stream of water through the sieve while brushing the residue gently with a soft camel-hair brush until all of the sample that will pass through the sieve has passed through it. Be sure that

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all of the lead styphnate passing through the sieve is caught in a container and none is washed down the drain. At the completion of this operation, visually examine the screen of the sieve for the presence of lead styphnate.

4.5.6 Apparent density. Place 3 Ml of n-butyl alcohol into a 5 Ml graduate which has an internal diameter of about 7.5 mm and is graduated in 0.05 Ml divisions. Add 2 gm of the dry sample weighed to the nearest 1 mg. in small portions and wash down the walls of the graduate with n-butyl alcohol. To insure that all the sample is wetted, shake the graduate carefully by slowly inverting it at least five times. Adjust the volume by adding butyl alcohol to the 5 Ml mark. Allow the graduate to stand for 3 hours and note the volume occupied by the solid material. Calculate the apparent density as follows:

$$\text{Apparent density} = \frac{W}{V}$$

where: W = weight of sample, gm.
V = Observed volume, Ml

4.5.6.1 Apparent density (alternate method). Transfer approximately 2 to 3 grams of the wet sample obtained in 4.4.3.1 into a Buchner funnel fitted with medium porosity filter paper. Rinse the wet sample with denatured ethyl alcohol and displace water. Rinse the sample with n-butyl alcohol to displace the denatured ethyl alcohol. Transfer the sample to a 5 or 10 Ml graduate, having 0.1 Ml graduations. Rinse down the sides of the cylinder with enough n-butyl alcohol to slurry the sample. Stir with a wooden stick, or agitate by some means in order to free any trapped air bubbles. Fill to the top cylinder graduation with n-butyl alcohol as accurately as possible, being sure the sides are rinsed clean. Weigh the cylinder and contents to the nearest mg. Record this weight as W1. Allow the cylinder contents to settle for a minimum of 2 hours, then note volume occupied by the sample. Record this volume as V. Determine the weight of the cylinder filled to the top cylinder graduation with n-butyl alcohol only (no sample) to the nearest 1 mg. Record this weight as W2.

Calculate the apparent density as follows:

$$\text{Apparent density} = \frac{D_s(W_1-W_2)}{V(D_s-D_b)}$$

where:

Ds = Density of normal lead styphnate
Db = Density of N-butyl alcohol

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4.5.7 Acidity. Transfer 1 gm of the dry sample weighed to the nearest 1 mg to a 150 Ml beaker. Add 100 Ml of freshly boiled and cooled distilled water having a pH of 6.0 + 0.5 and allow the mixture to stand with occasional stirring for 15 minutes. Filter and determine the pH of the filtrate at 25° + 2°C by means of a pH meter that is capable of measuring pH within 0.1 pH unit. The pH meter shall have glass and calomel electrodes and shall be calibrated with standard buffer solution. Test another 100 Ml portion of the water concurrently with the sample solution and correct the pH of the sample solution for any change in pH found in this blank solution.

4.5.7.1 Acidity (alternate method). Transfer approximately 50 Ml of the water covering the sample obtained in 4.4.3.1 to a 200 Ml beaker. Determine the pH value of this solution in accordance with method given in ASTM - E70.

4.5.8 Material insoluble in ammonium acetate. Transfer 1 gm of the dry sample, weighed to the nearest 1 mg, to a 250 Ml beaker. Add 100 Ml distilled of water and then add with constant stirring slowly 100 Ml of 20 percent ammonium acetate solution. Allow to stand for 15 minutes with frequent stirring and filter the solution through a tared sintered-glass crucible of fine porosity. Transfer the residue to the crucible with distilled water and wash the crucible with distilled water until the washings are colorless. Dry in an oven at 100° to 105°C for 1 hour, cool in a desiccator, and weigh. Calculate the percent material insoluble in ammonium acetate as follows:

$$\text{Percent material insoluble in ammonium acetate} = \frac{100A}{W}$$

where: A = weight of residue, gm.
W = weight of sample, gm.

4.5.9 Material soluble in ether. Transfer 2 gm of the dry sample weighed to the nearest 1 mg to a 150 Ml beaker, add 50 Ml of anhydrous ethyl ether and allow to stand with occasional stirring for about 15 minutes. Filter through a dry No. 40 Whatman filter paper or equivalent and catch the filtrate in a tared 100 Ml beaker. Wash with a total of 20 Ml of ether. Evaporate off the ether in the 100 Ml beaker on a steam bath or by means of a current of dry air. Place in a vacuum desiccator containing sulfuric acid until constant weight to the nearest 0.1 mg is obtained. Run a blank determination of 70 Ml of ether. Calculate the percent material soluble in ether as follows:

$$\text{Percent material soluble in ether} = \frac{100\ (A-B)}{W}$$

where: A = weight of residue from sample, gm.

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B = weight of residue from blank, gm.
W = weight of sample, gm.

4.5.10 Lead. Transfer 0.45 to 0.55 gm of the dry sample weighed to the nearest 0.1 mg to a 250 Ml beaker. Add 10 Ml of concentrated nitric acid and 10 Ml of concentrated sulfuric acid. Cover the beaker with a watch glass and heat at moderate temperature on a hot plate until dense fumes of sulfur trioxide are evolved. (Oxides of sulfur and nitrogen are toxic, precautions should be taken to assure personnel are not exposed to these fumes.) Allow the beaker to cool for about 2 minutes, add 5 Ml of concentrated nitric acid, and heat again until dense fumes of sulfur trioxide are evolved. Repeat the addition of nitric acid and heating to fumes until all the organic matter is destroyed and the solution is practically colorless. Remove the beaker from the hot plate and allow it to cool. Wash down the watch glass and side walls of the beaker with about 10 Ml of distilled water. Place the uncovered beaker on the hot plate and again heat until dense fumes of sulfuric acid are evolved. Allow to cool, then cautiously dilute to about 100 Ml with distilled water and again allow to cool. Add 25 Ml of 95 percent ethyl alcohol and let it stand for at least 2 hours. Filter through tared sintered porcelain crucible of fine porosity. Transfer the precipitate to the crucible with 5 percent sulfuric acid and twice with 95 percent ethyl alcohol. Dry in an oven at 100° to 105°C for 1/2 hour and then ignite in a muffle furnace at 550° to 600°C for 15 minutes. Cool in a desiccator and weigh to the nearest 0.1 mg. Calculate the percent lead in the sample as follows:

$$\text{Percent lead} = \frac{68.32A}{W}$$

where: A = weight of ignited residue, gm.
W = weight of sample, gm.

4.5.11 Nitrogen.

4.5.11.1 Apparatus. Storage and titration system are as shown in figures 1 and 2.

4.5.11.2 Reagents.

4.5.11.2.1 Potassium dichromate solution (0.2000 N). Dry National Bureau of Standards potassium dichromate NBS Standard 136 in an oven at 130°C for 2 hours, cool, dissolve 9.8070 gm in distilled water, and dilute to 1 liter in a volumetric flask. This is a primary standard.

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4.5.11.2.2 Acetic acid solution (25 percent). Add 250 Ml of glacial acetic acid to 750 Ml of distilled water.

4.5.11.2.3 Titanous chloride solution (0.2N). This solution can be prepared from either 20 percent titanous chloride solution or from titanium hydride.

4.5.11.2.3.1 Preparation from 20 percent titanous chloride solution. Filter about 150 Ml of 20 percent titanous chloride solution through a glass one filtering crucible. Add 100 Ml of concentrated hydrochloric acid and mix by means of a current of inert gas (oxygen-free carbon dioxide or nitrogen). Add 750 Ml of oxygen-free distilled water and again mix by means of a current of inert gas. Store in an amber bottle under carbon dioxide supplied by a Kipp generator as shown in figure 1. The solution need not be standardized, as it is compared with the standard ferric ammonium sulfate each time an analysis is made.

4.5.11.2.4 Ammonium thiocyanate solution (20 percent). Dissolve 20 gm of NH_4CNS in 80 gm of distilled water.

4.5.11.2.5 Sodium acetate solution (20 percent). Dissolve 20 gm of CH_3COONa in 80 gm of distilled water.

4.5.11.2.6 Stannous chloride solution. Dissolve 50 gm of $SnCl_2\ 2H_2O$ in 100 Ml of concentrated hydrochloric acid and dilute to 1 liter with distilled water.

4.5.11.2.7 Diphenylamine sulfonate indicator solution. Add 0.095 gm of the barium salt of diphenylamine sulfonic acid to 100 Ml of distilled water and stir until solution is complete. Add 10 Ml of dilute sulfuric acid (1 to 1), let stand several hours, and filter. Add 500 Ml of 85 percent phosphoric acid and dilute to 1 liter.

4.5.11.2.8 Ferric ammonium sulfate solution (0.2N). Dissolve 100 gm of ferric ammonium sulphate $(FeNH_4(SO_4)_2\ 12H_2O$ in one liter of 5 percent sulfuric acid. The normality is determined as follows: Pipet 25 Ml of the ferric ammonium sulfate solution into a 500 Ml wide mouth Erlenmeyer flask and add 25 Ml of concentrated hydrochloric acid. Heat nearly to boiling point and add the stannous chloride solution dropwise and with stirring until the yellow color of ferric chloride is discharged and then add 1 drop stannous chloride solution in excess (no more). Cool and add 10 Ml of a saturated mercuric chloride solution. Dilute to about 200 Ml and let stand for 5 minutes with occasional shaking. Add 15 Ml diphenylamine sulfonate indicator solution and titrate with a standard 0.2000 N potassium dichromate solution. The change at the end point is

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from light green to purple. The normality of the ferric ammonium sulfate solution is calculated as follows:

$$\text{Normality of ferric ammonium sulfate solution} = \frac{A \times N}{B}$$

where: A = potassium dichromate solution, Ml.
N = normality of potassium dichromate solution.
B = ferric ammonium sulfate solution, taken by titration, Ml.

4.5.11.3 Procedure. Transfer 0.6 to 0.8 gm of the dry sample weighed to nearest 0.1 mg to a 100 Ml beaker. Add 50 Ml of 25 percent acetic acid solution and stir to dissolve. Transfer to a 250 Ml volumetric flask and dilute to the mark. Attach the source of inert gas (oxygen-free carbon dioxide or nitrogen) to the titration flask as shown in Figure 1 and allow the gas to flow for 5 minutes to displace the air. Continue the flow of gas during the entire titration procedure. Pipet a 25 Ml aliquot of the solution of the sample into the titration flask (figure 2). Add 30 Ml of 20 percent sodium acetate solution and 20.00 Ml of 0.2 N titanous chloride solution and swirl the flask for about 20 seconds. Add 25 Ml of 15 percent hydrochloric acid. Titrate with ferric ammonium sulfate solution until the purple color begins to fade, then add 5 Ml of 20 percent ammonium thiocyanate solution and continue the titration to a faint permanent pink. A blank run should be made using the quantities of reagents as above, but omitting the sample. This will provide a value for the amount of ferric ammonium sulfate equivalent to the 20.00 Ml of titanous chloride solution. Calculate the percent nitrogen in the sample as follows:

$$\text{Percent nitrogen} = \frac{(A-B) \times 0.00233N \times 1000}{W}$$

where: A = ferric ammonium sulfate solution for the blank, Ml.
B = ferric ammonium sulfate solution for the sample, Ml.
N = Normality of ferric ammonium sulfate solution.
W = weight of sample, gm.

4.5.11.4 Nitrogen (alternate method). Determine the nitrogen content of the lead styphnate in accordance with Method 409.1 of MIL-STD-650.

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4.5.12 Instantaneous flash point. Use samples of the size that may be held on about 1/16 inch of the flat end of an ordinary toothpick and a heated Dennis melting point bar. The instantaneous flash point is determined by noting the times required for the explosion of samples dropped on the bar at several temperatures slightly lower than the instantaneous flash point. By plotting a time-temperature curve and extrapolating to zero time, the instantaneous flash point is obtained.

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Figure 1. Storage System, Figure 2. Titration Flask

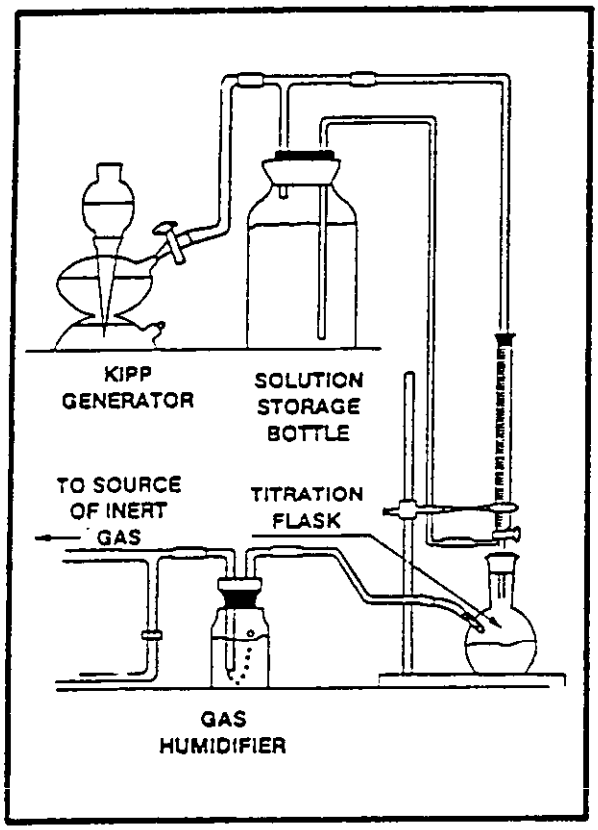


FIGURE 1. Storage System

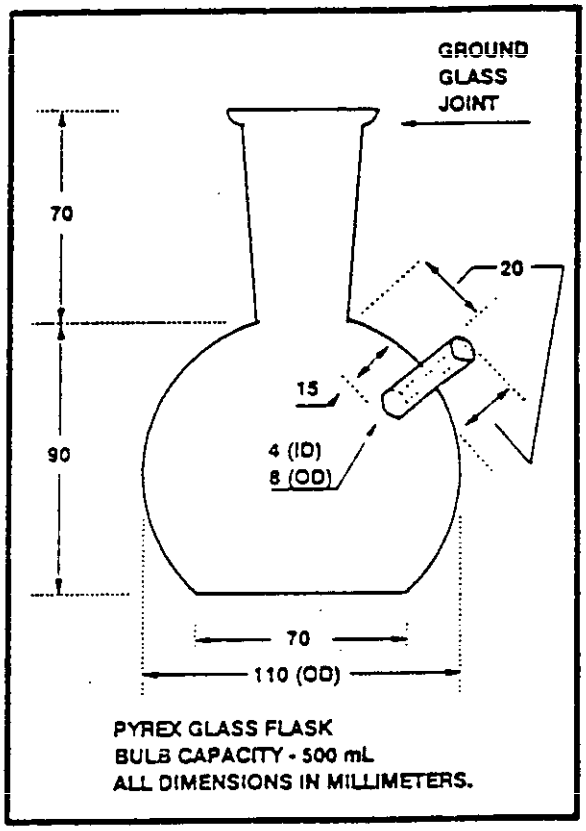


FIGURE 2. Titration Flask

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5. PACKAGING

5.1 Unit packaging

Lead Styphnate, normal shall be packaged level A, B or C as required by procuring activity, in accordance with MIL-STD-2073-1, meeting the requirements of the Code of Federal Regulations, Title 49 (49 CFR).

5.2 Intermediate pack (Not Applicable)

5.3 Exterior pack

No more than 150 pounds of dry Lead Styphnate, normal packaged Level A, B or C as applicable shall be overpacked in accordance with MIL-STD-2073-1, meeting the requirements of 49CFR.

5.4 Marking

5.4.1 Unit packaging

The Level A, B or C unit packages shall be marked in accordance with MIL-STD-129-1 and the following: NOMENCLATURE and QUANTITY, NATIONAL STOCK NUMBER (OR PART NUMBER), LOT NUMBER

5.4.2 Intermediate pack (Not Applicable)

5.4.3 Exterior pack

Item Identification marking for level A, B, or C exterior pack shall be in accordance with MIL-STD-129-1 and as required in 49 CFR, part 172, subpart D, Paragraph 172.301 (A) and the following:

NOMENCLATURE AND QUANTITY
NATIONAL STOCK NUMBER
LOT NUMBER
GROSS WEIGHT
ANY OTHER SPECIAL MARKING
POP MARKING (see note)

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5.5 Performance oriented packaging (POP). Packaging and packing as specified above, shall be in accordance with the applicable packing method as specified in the Code of Federal Regulations, Title 49, (49 CFR). Further, it shall be demonstrated that the pack meets all of the POP test requirements in accordance with 49 CFR. A POP test report shall be generated in accordance with DI-PACK-81059 (Data Item Description) following the tests. POP testing may be waived if an acceptable analogy in accordance with 49 CFR, to another pack which has successfully completed the testing, can be made. An analogy must also be documented in accordance with DI-PACK-81059. When completed, either POP test report must be kept on file by the contractor. The POP marking to be applied to the exterior pack shall be as specified by the government after review and acceptance of the submitted POP test reports.
NOTE: After completion of the requirements of 5.5, POP marking shall be applied to the pack as specified by the Government.

5.6 Item hazard classification. All U.S. manufacturers shall make certain that the item is tested in accordance with part 173, Subpart C, Section 173.58(A) of 49 CFR Parts 100-180 to assign proper Class and Division for all explosives (Division 1.1, 1.2, 1.3 and 1.4 Explosives). Registration with the Associate Administrator of Hazardous Materials Safety is required in accordance with Part 173, Subpart C, Section 173.56(B) (1) or 173.56(C) of 49 CFR so that markings in accordance with 172, Subpart D, Section 172.301(A) and 172.320(A) are met.

All foreign manufacturers shall make certain that the dangerous goods are tested in accordance with United Nations Committee of Experts on the Transportation of Dangerous Goods (as published in UN Document ST/SG/AC.10.11, latest revision, Recommendations on the Transportation of Dangerous Goods - Test and Criteria) to determine the proper division (Class 1-9 and Division 1.1-1.6 for explosives). Registration for air and vessel transport is required for each manufacturing country's National Competent Authority. The Hazard Classification Letter of Competent Authority is issued in accordance with part 2, paragraph 1.3 of the International Civil Aviation Organization (ICAO) Technical Instructions and approves the hazardous classification and compatibility group assignment and assigns the appropriate shipping name to the dangerous goods. The proper packaging, marking and labeling is contained in the United Nations Committee of Experts on the Transportation of Dangerous Goods (as published in UN Document ST/SG/AC.10.1, latest revision, Recommendations on the Transport of Dangerous Goods).

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For air transport, the dangerous goods must comply with the provisions of the International Air Transport Association (IATA) Dangerous Goods Regulations and for vessel transport, the dangerous goods must comply with the provisions of the Intergovernmental Maritime Organization's International Maritime Dangerous Goods Code (IMDG Code).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Normal lead styphnate, lead 2,4,6 tri nitro resorcinate ($C_6H(NO_2)_3O_2Pb \cdot H_2O$), described in this specification is an explosive intended for use in priming compositions. Normal lead styphnate is classified as a high explosive. (All necessary precautions in the manufacture, handling and storage of such materials should be observed.)

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this detail specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. First article requirements (see 3.3, 4.3 and 6.4).

d. Level of packing protection required.

6.3 Waived requirements. The requirements for crystal dimensions (see 3.1.3) and instantaneous flash point (see 3.2, Table I) may be waived at the discretion of the purchasing agency.

6.4 First article. When first article inspection is required the contracting officer should specify that the contractor submit a first article sample as specified in 4.3.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the

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bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Submission of inspection equipment designs for approval. Submit contractor equipment designs as required to: Commander, U. S. Army ARDEC, ATTN: SMCAR-QAR-R, Picatinny Arsenal, NJ 07806-5000 (see MIL-A-48078). This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5.1 Submission of contractor test. The POP tests must be submitted to the Commander, U.S. Army ARDEC, ATTN: SMCAR-AEP, Picatinny Arsenal, New Jersey 07806-5000. (Note: If a POP test report is prepared against an acceptable analogy, the analogy POP test report must also be submitted to SMCAR-AEP. The IATA and IMDG documents shall be forwarded to the U.S. Army Armament Munitions and Chemical Command (AMCCOM), ATTN: AMSMC-PC, AMSMC-PG and AMSMC-SFS, Rock Island, Illinois 61299-6000).

6.6 Equivalent test methods. The contractor may request to use alternate test method(s) providing that the proposed method is as a minimum equivalent (accuracy and precision) to the method given in this specification. Prior approval of the contracting officer is required for use of equivalent test methods. A description or the proposed method should be submitted through the contracting officer to Commander, SMCAR-QAR-R (D), Picatinny Arsenal, NJ 07806-5000. This description should include the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.7 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.8 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval by the technical activity responsible for the preparation of this specification.

6.9 Subject term (key word) listing. Explosive, primer, lead 2, 4, 6 tri nitro resorcinate.

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6.10 Change from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army (AR)

Preparing activity:
Army-AR

(Project 1376-A494)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. **The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.**
2. **The submitter of this form must complete blocks 4, 5, 6, and 7.**
3. **The preparing activity must provide a reply within 30 days from receipt of the form.**

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-00757B (AR)	940516

3. DOCUMENT TITLE: LEAD STYPHNATE, NORMAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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